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n8n Automation Mastery
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MENTOR COPY

Module 1 · Automation Foundations

DAY 03 of 25

n8n Data Model: Items, JSON & Binary

"Jo data samajh gaya, woh n8n samajh gaya."

⚡ AAJ HUM YEH BANAYENGE

CSV Sales → City-wise Report Email

Form par sales ki CSV file upload karo. Workflow file padhta hai, city-wise total nikaalta hai aur ek saaf HTML table email karta hai.

Split Out | Aggregate | Summarize | Extract from File

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

n8n mein sab kuch "items" hai. Jo student items ko samajh gaya, usse koi bhi workflow mushkil nahi lagega.

🎯 Aaj ke baad aap yeh kar paoge

01 JSON basics: objects, arrays, nesting

🌍 Visiting card: Naam, Phone, Email — har field ka ek label aur value.
⚙️ {"name": "Aman", "phones": ["98...", "99..."], "address": {"city": "Pune"}}

02 n8n items — har node items ki list process karta hai

🌍 Conveyor belt par 10 boxes — machine har box par same kaam karti hai.
⚙️ HTTP se 10 users aaye → Edit Fields har user par chala → 10 output items.

03 Item linking & paired items

🌍 Order slip aur uska bill — dono ek doosre se jude hain.
⚙️ Paired items se pata rehta hai ki output item kis input item se bana.

04 Binary data: files, images, PDFs

🌍 Courier mein letter (JSON) + parcel attached (binary).
⚙️ Gmail attachment → binary.data → Extract from File → text.

05 Split Out, Aggregate & Summarize nodes

🌍 Mithai ka dabba kholna (split), wapas bharna (aggregate), ginti karna (summarize).
⚙️ Order ke line items Split Out → Summarize by product → Aggregate into one email.

06 Data pinning & mock data for testing

🌍 Rough copy par practice — final copy kharab nahi hoti.
⚙️ API ka output pin karo → aage ke nodes bina API call ke test karo.

⏱ Class ka plan (2 ghante)

10m	40m	40m	20m	10m

0:00-0:10 Recap

🪄 Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.

👩‍🏫 2 students se homework screen share karwao. Galti wala homework bhi dikhao — sabse zyada seekh wahin se aati hai.

0:10-0:50 Concept: Data Structure

Visual walkthrough of items flow

🪄 Teacher ke paas 30 copies hain. Woh har copy par same kaam karti hai — check karna, marks dena. n8n bhi har item par wahi karta hai.

👩‍🏫 Pehle real-life example se shuru karo, phir n8n screen dikhao. Har concept ke baad 1 student se apne shabdon mein dobara bulwao.

0:50-1:30 Live Build

CSV upload → clean → split → aggregate report

🪄 Aaj ki kahani yaad hai — 'Tiffin factory ki assembly line'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.

👩‍🏫 Har 10 minute par poocho "sabka green tick aaya?". Jaan-boojh ke ek galti live hone do aur use debug karke dikhao.

1:30-1:50 **Quiz + Challenge**

Module 1 quiz (15 questions)

🪄 *Practice match — jitna yahan girenge, utna asli match mein kam girenge.*

👩‍🏫 Pehle 5 minute akele try karne do, phir hint do. Jo solve kare use screen share karwao.

1:50-2:00 **Q&A + Homework**

🪄 *Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?*

👩‍🏫 Chat se top 3 doubts live solve karo. Homework ka pehla step class mein hi karwa do, taaki koi blank na baithe.

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

JSON

Data likhne ka format: { "key": "value" }. Objects {} aur arrays [].

🪄 **Analogy:** Form ke fields aur unke jawab

CONCEPT 02

Items

Har node items ki LIST leta hai aur har item par kaam karta hai.

🪄 **Analogy:** Assembly line par boxes

CONCEPT 03

Split Out vs Aggregate

Split Out: 1 item ki list → kai items. Aggregate: kai items → 1 list.

🪄 **Analogy:** Packet kholna vs packet bharna

CONCEPT 04

Binary data

Files (PDF, image) JSON ke saath "binary" property mein chalti hain.

🪄 **Analogy:** Courier ke saath attached parcel

🔑 Aaj ka main idea

n8n mein data "items" ki list ke roop mein chalta hai. Har item ek JSON object hai — jaise ek form ka ek jawab. Agar 10 items aaye, to agla node 10 baar chalega. Yeh samajh aa gaya to n8n 50% aa gaya.

- Attendance register: har line = ek student = ek item
- Shopping list: har cheez = ek item
- WhatsApp group ke 20 messages = 20 items

📖 Ek kahani se samjho: Tiffin factory ki assembly line

1 📦 **Ek bada dabba aaya (CSV)**
= Extract from File

2 📤 **Dabba khola, 50 tiffin nikle**
= Split Out → 50 items

3 🧹 **Toote tiffin hataye**
= Filter

4 🏙️ **City-wise ginti**
= Summarize

5 🚚 **Report bhej di**
= Gmail

🪄 Ek line mein: “Teacher ke paas 30 copies hain. Woh har copy par same kaam karti hai — check karna, marks dena. n8n bhi har item par wahi karta hai.”

🗺️ Aaj ka workflow: CSV Sales → City-wise Report Email

Data left se right chalta hai. Har dabba ek n8n node hai.



💰 Isse kitna kama sakte ho?

Distributors, D2C brands (daily/weekly sales report) — aam taur par **₹5,000 – ₹12,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1 ⚡ **Manual Trigger**
Testing ke liye

2 🌐 **HTTP Request**
CSV file URL se download (Response: File)

3 📄 **Extract from File**
Operation: Extract from CSV → har row ek item

4 🔀 **Filter**
amount > 0 (khaali/galat rows hatao)

5 📈 **Summarize**
Group by: city, Sum: amount, Count: orders

6 📧 **Gmail**
City-wise summary table email

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- ☐ Connect: **Gmail (Google OAuth)**
- ☐ Sample CSV download karo (neeche link) — columns: order_id, city, amount

✅ Test kaise karein

1. Form ka Test URL kholo → sample-sales.csv upload karo
2. Har node ka output dekho: items kaise group hue
3. Email mein table check karo

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😩 Pehle (manual)

Excel kholna	2 min
City-wise filter + total	20 min
Summary type karna	10 min
Email bhejna	3 min
	35 min

⚡ Ab (n8n)

CSV aate hi summary email

0 min

⏱ Har report par 35 min bache, calculation galti zero.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo — 📋 Select karke copy karo

```
{{ $input.all().length }}  // kitne items aaye
```

📦 Sample data — input se output tak

📥 Input — 📋 Select karke copy karo

```
[
  {
    "city": "Delhi",
    "amount": 1200
  },
  {
    "city": "Pune",
    "amount": 800
  },
  {
    "city": "Delhi",
    "amount": 500
  }
]
```

📤 Output — 📋 Select karke copy karo

```
[
  {
    "city": "Delhi",
    "sum_amount": 1700,
    "count": 2
  },
  {
    "city": "Pune",
    "sum_amount": 800,
    "count": 1
  }
]
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude — 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Form par sales ki CSV file upload karo. Workflow file padhta hai, city-wise total nikaalta hai aur ek saaf HTML table email karta hai.

Flow: Form (file upload) (sales.csv) → Extract from File (CSV → items) → Code (City-wise sum) → Code (HTML table) → Gmail (Report email)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 Debug helper prompt — 📋 Select karke copy karo

```
I'm building the n8n workflow "CSV Sales → City-wise Report Email" (nodes: Upload Sales CSV, Read CSV, City-wise Total, Build HTML Table, Email Report).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ Workflow JSON
5 nodes · ready to import
`adifyskills.com/workflows/day-03-csv-city-sales-report.json`

🌐 Lesson page
Copy buttons, quiz aur video links
`adifyskills.com/day/3`

🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Gmail (Google OAuth)" connect karo.
Webhook/Form test par kuch nahi ho raha?	"Listen for test event" / Test URL use karo aur 2 minute ke andar request bhejo. Production URL sirf Active workflow par chalta hai.

⚠️ Yeh galtiyan sabse common hain

! Ek item expect karna jab 50 items aa rahe hon (har email 50 baar jayega!)
! Nested array ko bina Split Out ke process karna
! Binary property ka naam galat dena (data vs file)

⭐ Pro tips — time bachane wali tricks

★ Output panel mein Table / JSON / Schema views badal ke dekho
★ Item count node ke upar dikhta hai — hamesha check karo
★ Aggregate se pehle Sort karo agar order matter karta hai

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

CSV sales file ko process karke city-wise summary nikaalna aur email karna.

📝 Homework

Public API (e.g. JSONPlaceholder) se data laake usse Split Out + Aggregate se transform karo.

☐ Homework ho gaya

BEGINNER
Sirf Delhi ke orders filter karo ☐

INTERMEDIATE
Top 3 cities sort + limit ☐

ADVANCED
Summary ko Excel file bana ke attach karo ☐

🧩 Aaj ka challenge — pehle khud try karo

🎯 Report mein sirf woh cities dikhao jinki sales ₹5,000 se zyada hai.

💡 Hint 1
Code node ke baad ek naya node lagao jo items ko chhaant de.

💡 Hint 2
Filter node mein condition: sales (number) "is greater than" 5000.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. 10 items ek Gmail node mein jaayein to kitne emails?

- (A) Error
- (B) 1
- (C) 10
- (D) 0

Q2. Kai items ko ek list banane ka node?

- (A) Filter
- (B) IF
- (C) Split Out
- (D) Aggregate

Q3. PDF file n8n mein kahan hoti hai?

- (A) binary
- (B) headers
- (C) credentials
- (D) json

Q4. JSON mein list kis bracket mein?

- (A) {}
- (B) []
- (C) ()
- (D) <>

📖 Aaj ke naye shabd

Workflow
Steps ki ek chain jo apne aap chalti hai — jaise recipe ke steps.

Node
Workflow ka ek dabba/step. Har node ek kaam karta hai (email bhejna, data laana...).

Trigger
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

JSON
Data likhne ka format: { "naam": "Ravi", "city": "Delhi" } — key aur value ke jode.

Item
Data ki ek row/record. n8n mein data items ki list ki tarah flow karta hai.

Binary
File wala data — PDF, image, CSV. Text se alag hota hai.

HTTP Request
Node jo kisi bhi API ko call kar sakta hai — n8n ka universal connector.

👨‍🏫 Mentor notes — class kaise lein

Yeh page sirf aapki copy mein hai. Students ki PDF mein nahi jaata.

- [ ] Whiteboard par items ko boxes ki tarah draw karo
- [ ] Ek hi workflow mein 1 item vs 10 items ka farak live dikhao
- [ ] Module 1 quiz yahin karwao (15 min)

🗣️ Har block ke liye mentor tip

10m — Recap

2 students se homework screen share karwao. Galti wala homework bhi dikhao — sabse zyada seekh wahin se aati hai.

Class se poochho: Kal wale workflow mein sabse mushkil step kaunsa laga? · Homework mein kahan error aaya aur kaise theek kiya? · Kal ka ek node naam batao aur woh kya karta hai?

40m — Concept: Data Structure

Pehle real-life example se shuru karo, phir n8n screen dikhao. Har concept ke baad 1 student se apne shabdon mein dobara bulwao.

Class se poochho: 5 items Gmail node mein jaayein to kitne emails jaayenge? · JSON mein list kis bracket mein hoti hai? · PDF file n8n mein kahan rehti hai?

40m — Live Build

Har 10 minute par poocho "sabka green tick aaya?". Jaan-boojh ke ek galti live hone do aur use debug karke dikhao.

Class se poochho: Is node ka output agle node mein kaise ja raha hai? · Agar yeh node fail ho to kya hoga? · Isko apne kaam/business mein kahan use kar sakte ho?

20m — Quiz + Challenge

Pehle 5 minute akele try karne do, phir hint do. Jo solve kare use screen share karwao.

Class se poochho: 10 items ek Gmail node mein jaayein to kitne emails? · Kai items ko ek list banane ka node? · PDF file n8n mein kahan hoti hai?

10m — Q&A + Homework

Chat se top 3 doubts live solve karo. Homework ka pehla step class mein hi karwa do, taaki koi blank na baithe.

Class se poochho: Aaj ka sabse useful node kaunsa laga? · Homework mein pehla step kya hoga? · Aaj ki class 1 line mein summarize karo.

✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 4

Expressions & Data Transformation

"{{ }} — n8n ki asli superpower."

☐ **Day 3 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

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🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → C
10
Har item ke liye node ek baar chalta hai.

Q2 → D
Aggregate

Q3 → A
binary

Q4 → B
[]

✅ Challenge ka solution

City-wise Total ke baad Filter node: {{ $json.sales }} → Number → is greater than → 5000. Phir Build HTML Table.

📚 Official docs

- Data structure ↗
- Binary data ↗
- Data pinning ↗
- Data mapping ↗
- Item linking ↗

▶ Videos

- n8n items data structure explained ↗
- n8n split out aggregate ↗

💬 Doubt poochho
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wa.me/918383891716

🌐 Day 3 lesson page
Quiz, copy buttons, videos
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